Department of Physics and Astronomy
First Year
2024-25

Tutorial Sheet 5

PHYS10302 Vibrations and Waves
PHYS10342 Electricity and Magnetism
PHYS10372 Mathematics 2
This sheet covers tutorials in weeks 6 of semester 2.

Problem solving can be a difficult skill to master. If you are finding it tough going, it may be helpful to try the following strategy:

1. Identify what the problem is about. Try to visualise the problem. Drawing a diagram may help.
2. Identify clearly what you are being asked to find.
3. What information have you been given? Sometimes even the simple task of writing down this information can help.
4. What (physics) principles and equations can you use?
5. How do you put points 1, 2, 3 and 4 together to get a solution? What maths will you need?
6. It is usually best to try and solve the problem algebraically, using symbols, even if you are given numerical values. This enables you to check at each stage and at the end, that you have correct and consistent dimensions. The last thing to do is to put numbers in, if a numerical answer is required.
7. Is the answer reasonable? If the answer is numerical, is the order-of-magnitude appropriate? Does your algebraic answer behave in the correct way when the variables take on very large or very small values, or in other limits?

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PHYS10302 Vibrations and Waves
Week 6 Questions

1. Calculate
   (a) the wavelength of middle C, frequency 256 Hz.
   (b) the wavelength of water waves traveling at 0.6 m/s and arriving on the water's edge at the rate of two every 3.0 s.
   (c) the frequency of red light of wavelength 700 nm.
   (d) the frequency of X rays of wavelength 0.1 nm.
   (e) the frequency of an AM radio station transmitting at 247 m.

2. (a) A ball of string weighs 0.30 kg and when unrolled is 20 m long. It is tied at one end, and a force of 25 N is applied at the other end. How long does it take waves to travel the length of the string?
   (b) A wire is of density 5.0×10^{-3} kg/m and waves travel in it at 20 m/s. Calculate the tension in the wire.

3. The equation of a transverse wave travelling along a string is given by

$$y(x,t) = 0.3 \sin[\pi(0.5x - 50t)],$$

   where y and x are in centimetres and t is in seconds.
   (a) Find the amplitude, wavelength, wave number, frequency, period and velocity of the wave.
   (b) Find the maximum transverse speed of any particle in the wave.

4. A wave of frequency $20\,\mathrm{s}^{-1}$ has a velocity of $80\,\mathrm{m\ s}^{-1}$.
   (a) How far apart are two points with a phase difference of 30° between their displacements?
   (b) At a given point, what is the phase difference between two displacements occurring at a time interval of 0.01 s?

5. A weight of mass M is hung at the bottom of a vertical steel wire of radius r. Find an expression for the speed of waves in the wire, and evaluate it for $M = 10$ kg, $r = 0.1$ mm.

   The ultimate strength of steel is quoted as the stress $S_{\max} = 10^{11}$ N/m^2. Find the maximum speed of waves in this steel wire.

   For this question take the density of steel as $\rho = 7.87 \times 10^3$ kg/m^3 and you can assume that steel behaves elastically right up to its breaking point.

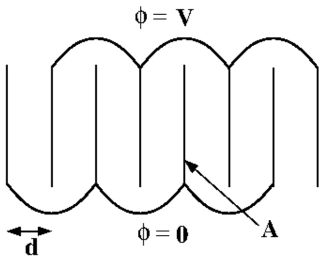
E&M Tutorial Sheet 5: Electric Energy and Dipoles

1) (Revision) Use Gauss' Law to calculate the electric field both between and outside a parallel plate capacitor where the plates have area *A*, charge *+Q* and *–Q* and are separated by *d*. Assume that the capacitor is "perfect" (i.e. *d* is small and the electric field between the plates is uniform). Also calculate the potential difference between the two plates and hence their capacitance.

2) Calculate the field in a parallel plate capacitor with plate area 1.00 m^2 and separation 1.00mm if the plates carry charges of + and – 1.00 µC. Hence calculate the energy density stored in the field and the total energy in the field.

3) The capacitor in Q2 is discharged through a resistor of 1.00kΩ. Calculate the current flowing at time *t* and hence the total energy dissipated from the resistor. Check that this agrees with the answer to Q2.

4) A 'capacitor stack' is constructed by stacking *N* parallel plates, each of area *A* and each separated by a distance *d*. Alternate plates are held at potentials zero and *V*. Calculate the total capacitance of such a stack.



5) Use the superposition principle to calculate the potential $\phi(r,\theta)$ due to charges *+q* and *–q* a distance *d* apart (i.e. an electric dipole *p=qd*). Use this result to calculate the electric field a distance *r>>d* from the dipole, both in the direction along the axis of the dipole and in the plane perpendicular to the dipole (and, if you are feeling confident with spherical polar coordinates by now, in the general direction). *The important point is that it is proportional to* $1/r^3$.

6) Summarise the important points from electrostatics in no more than one side of paper.

PHYS10372 Mathematics 2: Tutorial-05 Questions

These questions cover the material from week-5 of PHYS10372: the divergence theorem and line integrals.

For Q1-3: use the **divergence theorem** i.e. calculate the flux of a vector field $\boldsymbol{A}$ using the volume integral: $\int_V \boldsymbol{\nabla} \cdot \boldsymbol{A}\, dV$

1. Calculate the flux of the vector field: $xy^2\,\hat{\boldsymbol{\imath}} + yz^2\,\hat{\boldsymbol{\jmath}} + zx^2\,\hat{\boldsymbol{k}}$ out of the rectangular box: $0 \le x \le a,\ 0 \le y \le b,\ 0 \le z \le c$.
   [Answer: $\frac{abc}{3}(a^2 + b^2 + c^2)$]
2. Calculate the flux of the vector field: $r^2\,\hat{\boldsymbol{r}} + z^2\,\hat{\boldsymbol{k}}$ out of the quarter cylinder defined by $x^2 + y^2 = R^2, 0 \le z \le h$ and $0 \le y,\ 0 \le x$
   [Answer: $\frac{\pi}{2}\left(R^3 h + \frac{h^2 R^2}{2}\right)$]
3. Prove that the volume of an arbitrary shape is equal to one-third of the flux of the position vector: $\boldsymbol{r} = x\,\hat{\boldsymbol{\imath}} + y\,\hat{\boldsymbol{\jmath}} + z\,\hat{\boldsymbol{k}}$ out of the shape.

Line Integrals

4. An electric current, I, flows along the z-axis and gives rise to the magnetic field:

$$\boldsymbol{B} = \frac{\mu_0 I}{2\pi r}\,\hat{\boldsymbol{\theta}}$$

   where (r, θ, z) are cylindrical polar co-ordinates. Calculate the circulation ($\oint \boldsymbol{B} \cdot \boldsymbol{dl}$) of $\boldsymbol{B}$ around a square loop of side $2a$ which lies in the $z = 0$ plane and is centred at the origin.
   (You will need to express $\hat{\boldsymbol{\theta}}$ in terms of $\hat{\boldsymbol{\imath}}$ and $\hat{\boldsymbol{\jmath}}$)
   [Answer: $\mu_0 I$]
5. Calculate the anticlockwise circulation of the vector field: $\boldsymbol{A} = 5x\,\hat{\boldsymbol{\jmath}}$ around the circle $x^2 + y^2 = a^2,\ z = 0$ by direct evaluation (i.e. **not** using Stokes' Theorem).
   [Answer: $5\pi a^2$]
6. Calculate the anticlockwise circulation of the vector field: $\boldsymbol{B} = z^2\,\hat{\boldsymbol{\imath}} - y^2\,\hat{\boldsymbol{\jmath}}$ around a square of side a lying in the xz plane, two of whose edges lie along the x and z axes by direct evaluation (i.e. **not** using Stokes' Theorem).
   [Answer: a^3]